

Regression: Predicting House Prices

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Predicting house prices

How much is my house worth?



How much is my house worth?

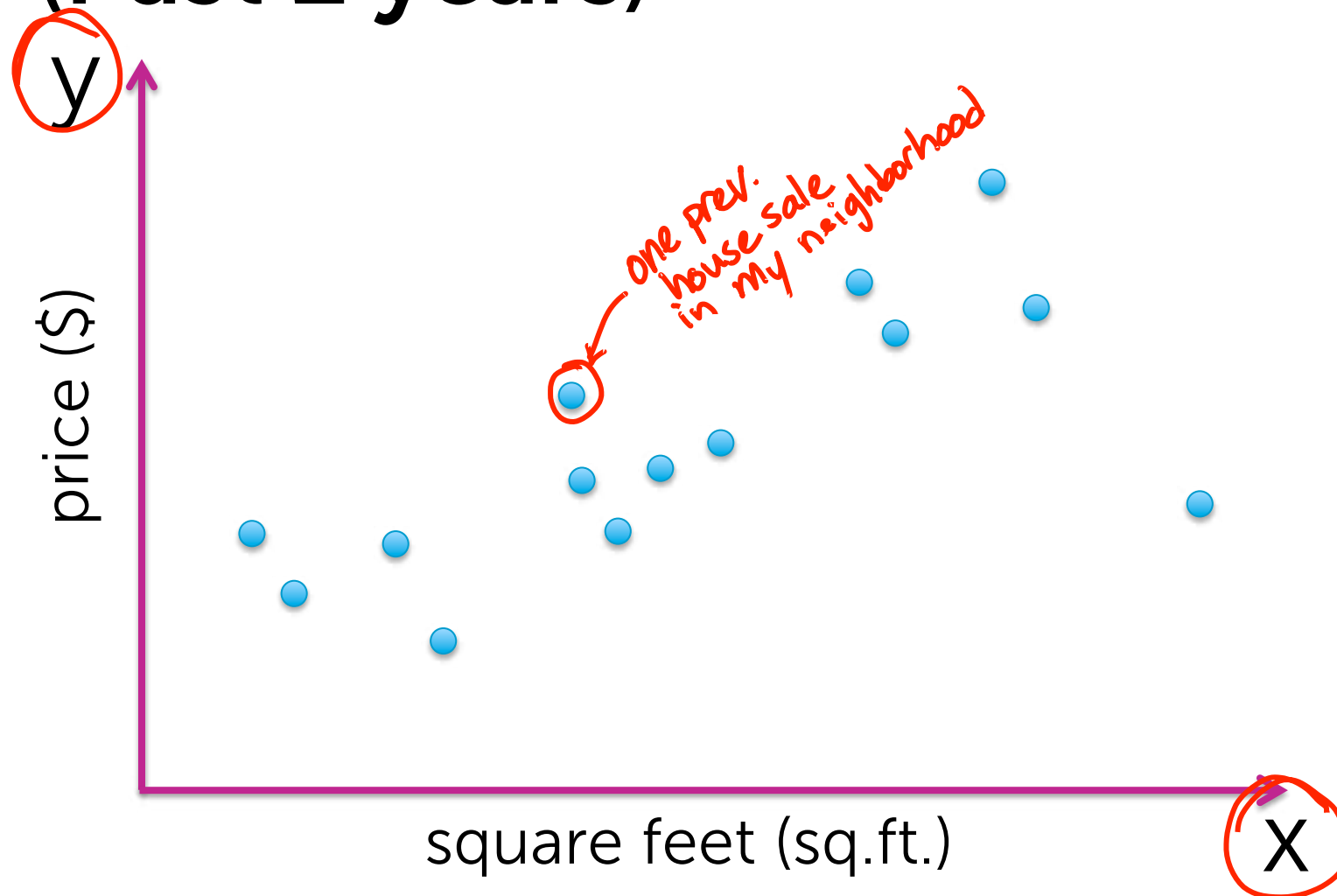


Look at recent sales in my neighborhood

- How much did they sell for?



Plot recent house sales (Past 2 years)

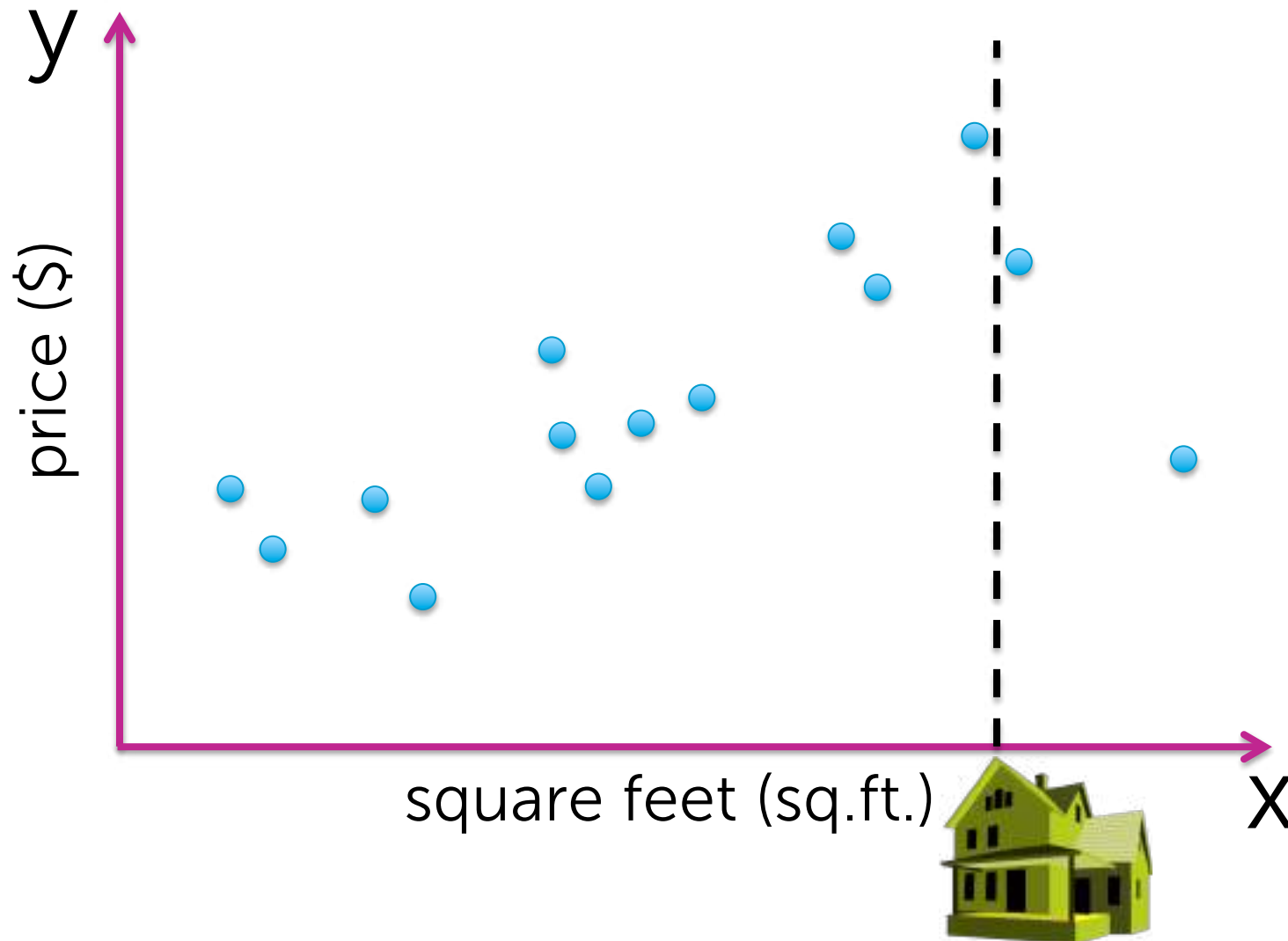


Terminology:

x – feature,
covariate, or
predictor

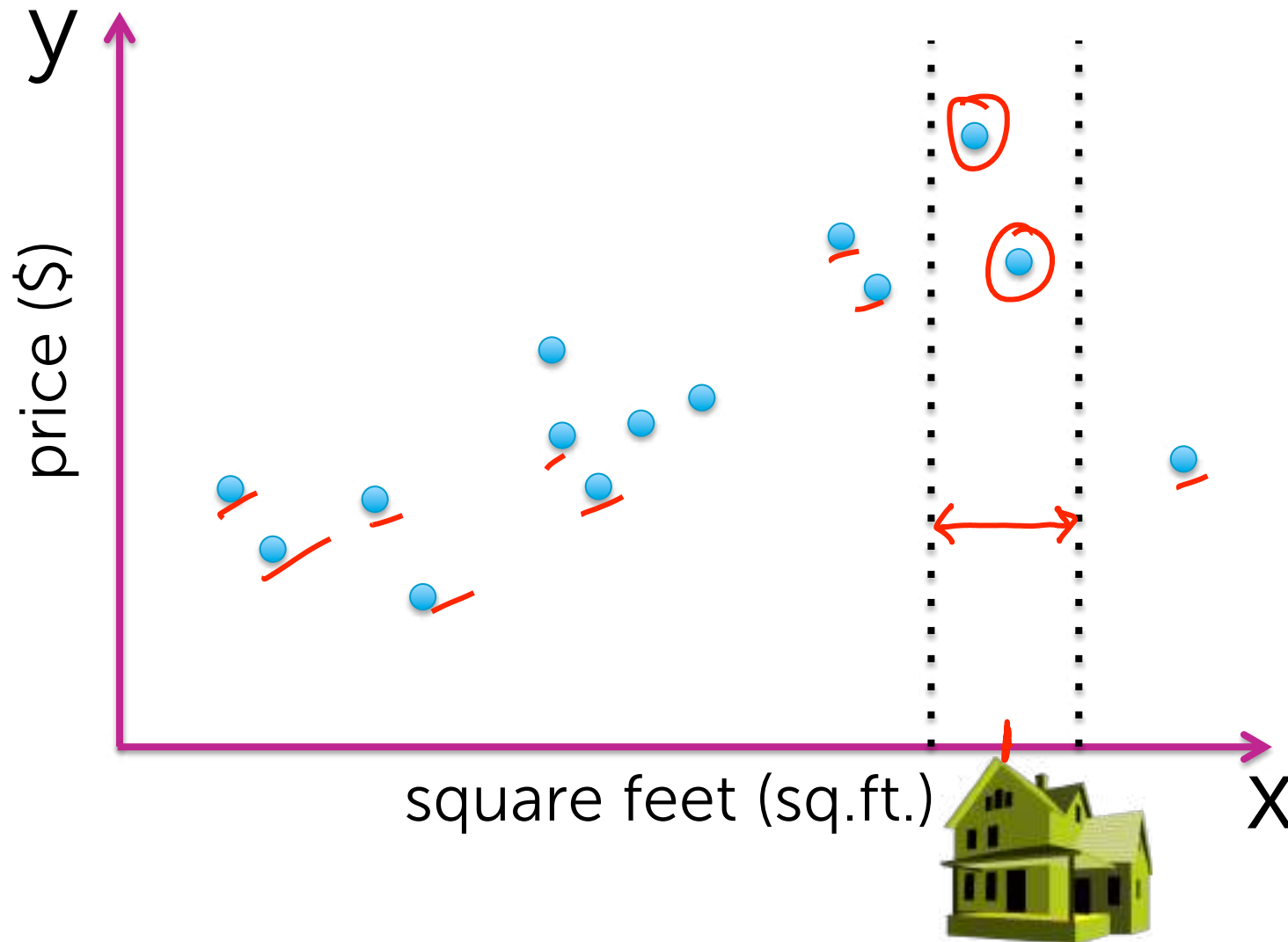
y – observation or
response

Predict your house by similar houses



No house sold recently had *exactly* the same sq.ft.

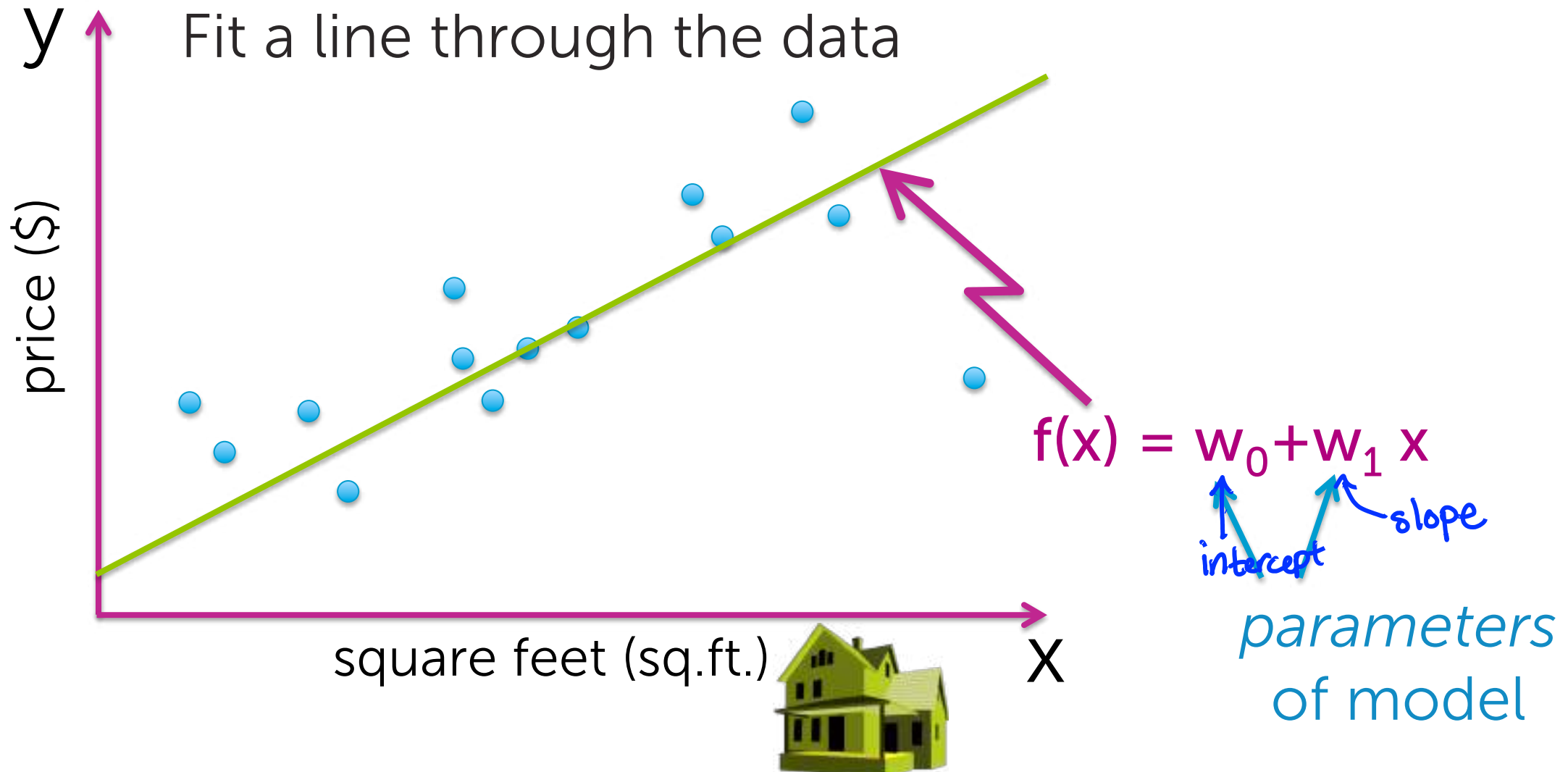
Predict your house by similar houses



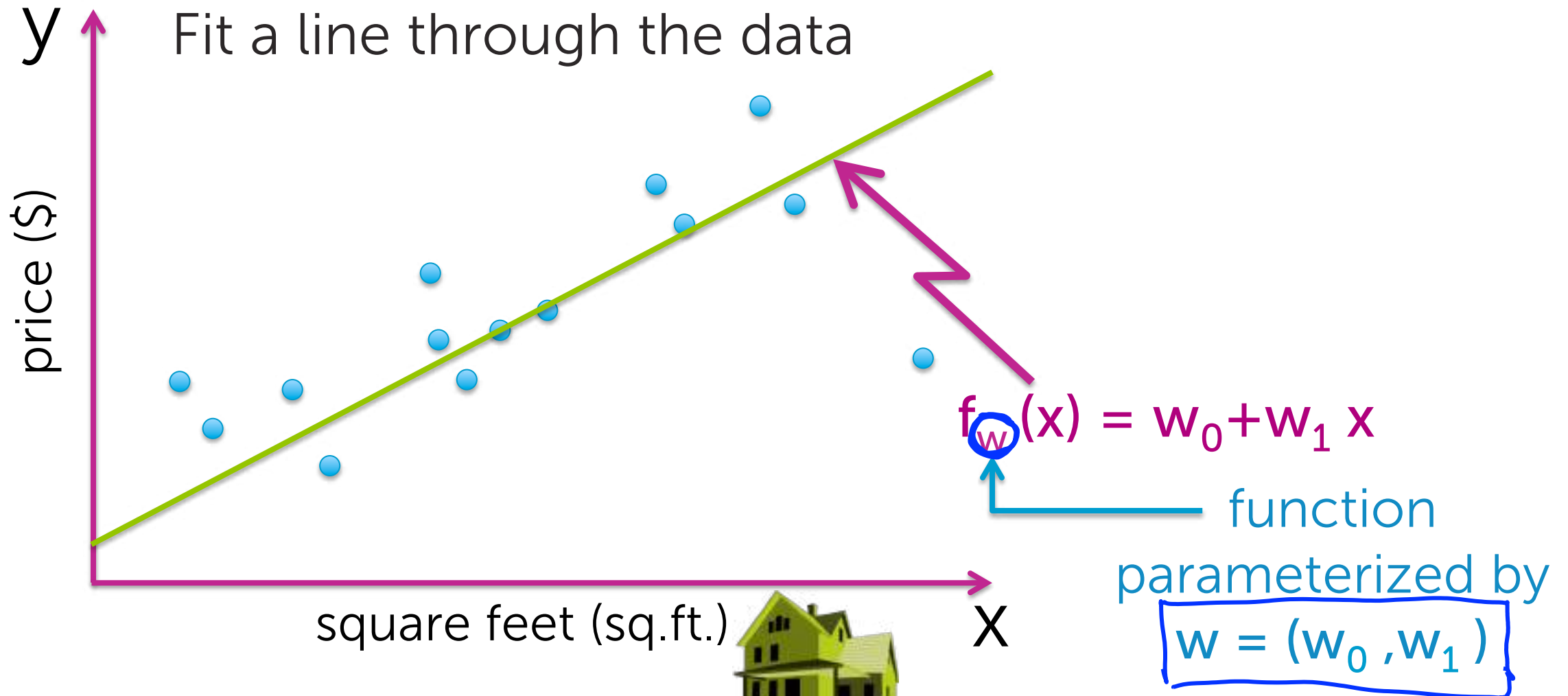
- Look at average price in range
- **Still only 2 houses!**
- Throwing out info from all other sales

Linear regression

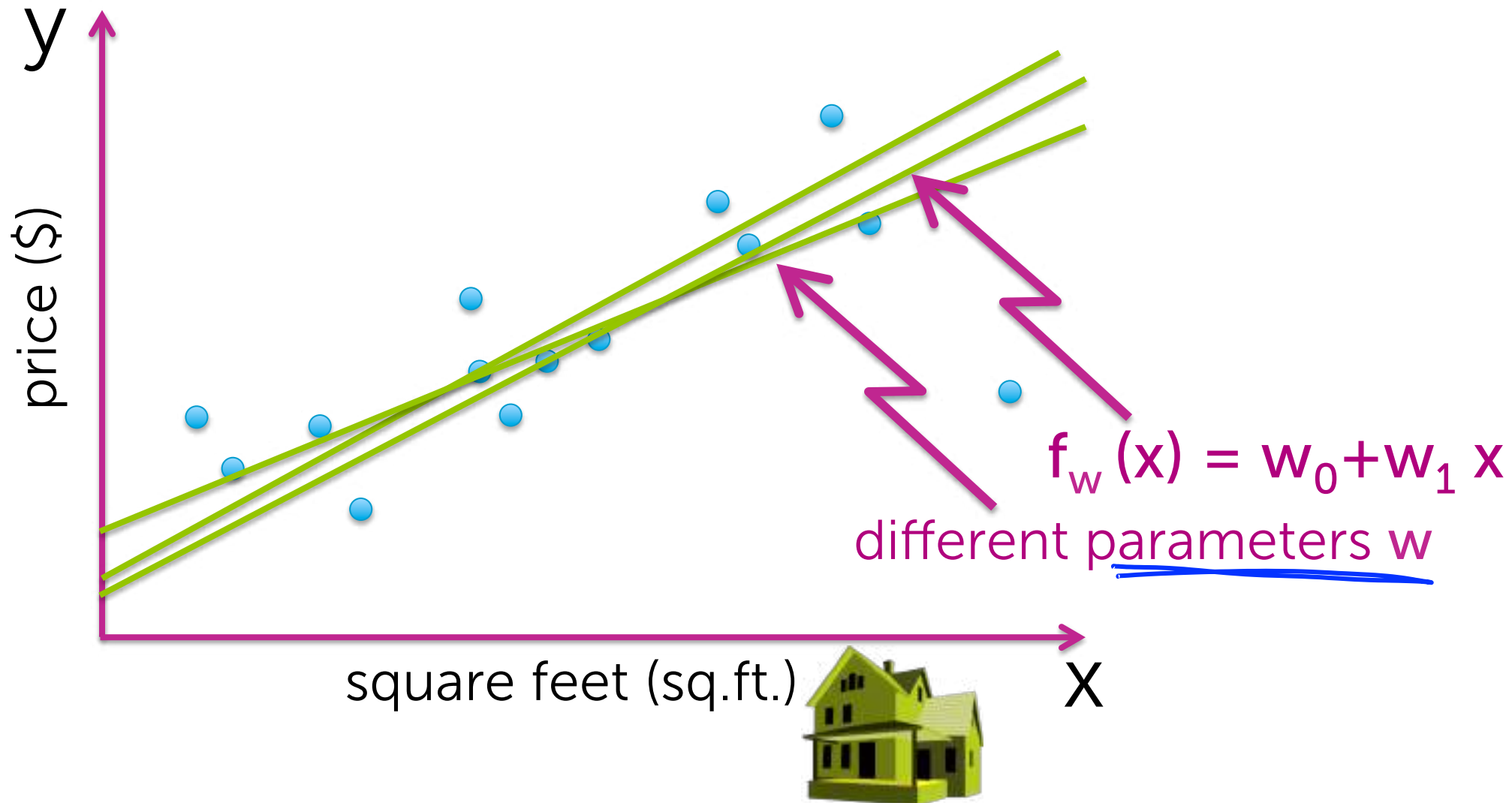
Use a **linear** regression model



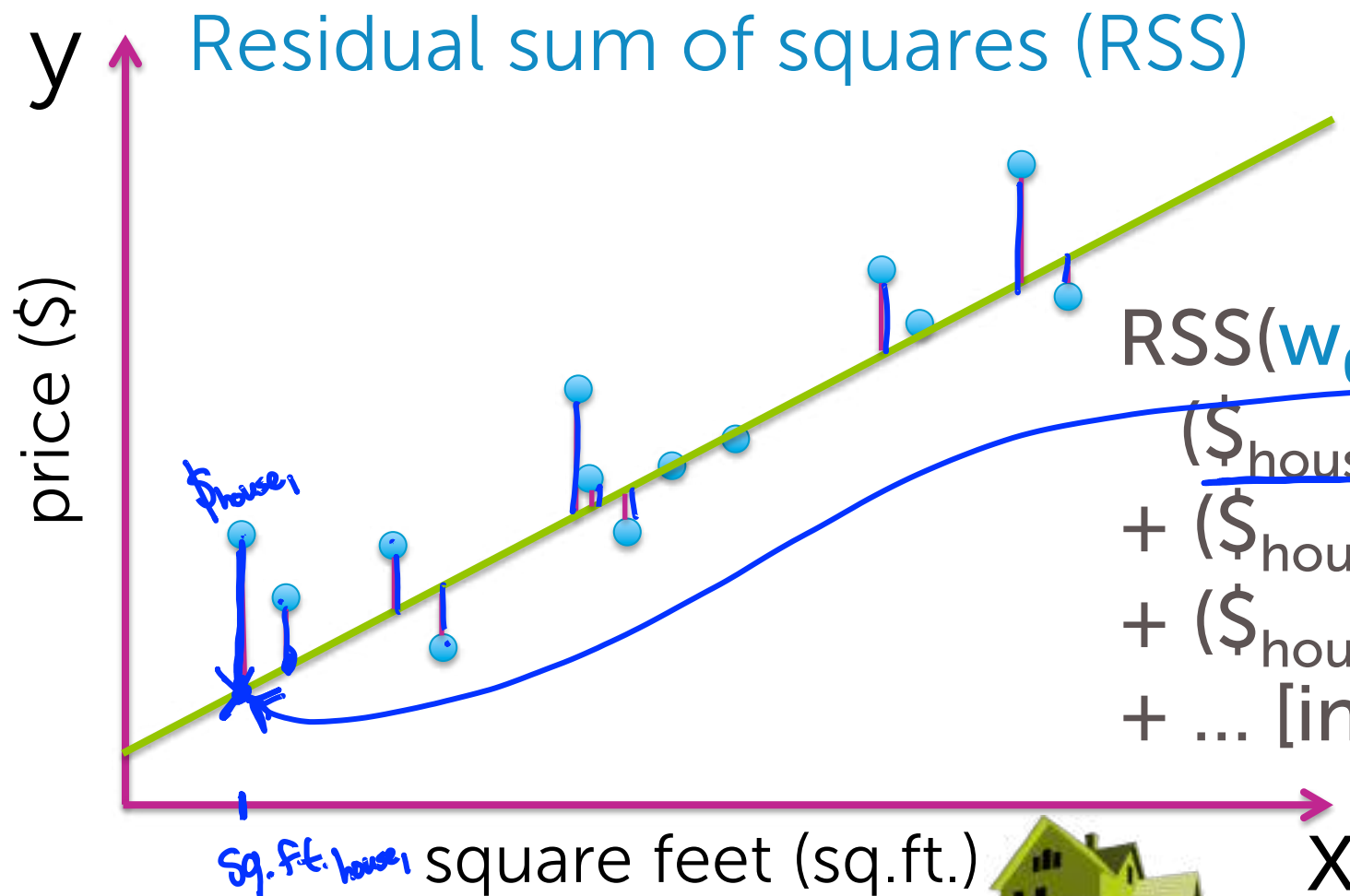
Use a **linear** regression model



Which line?



"Cost" of using a given line

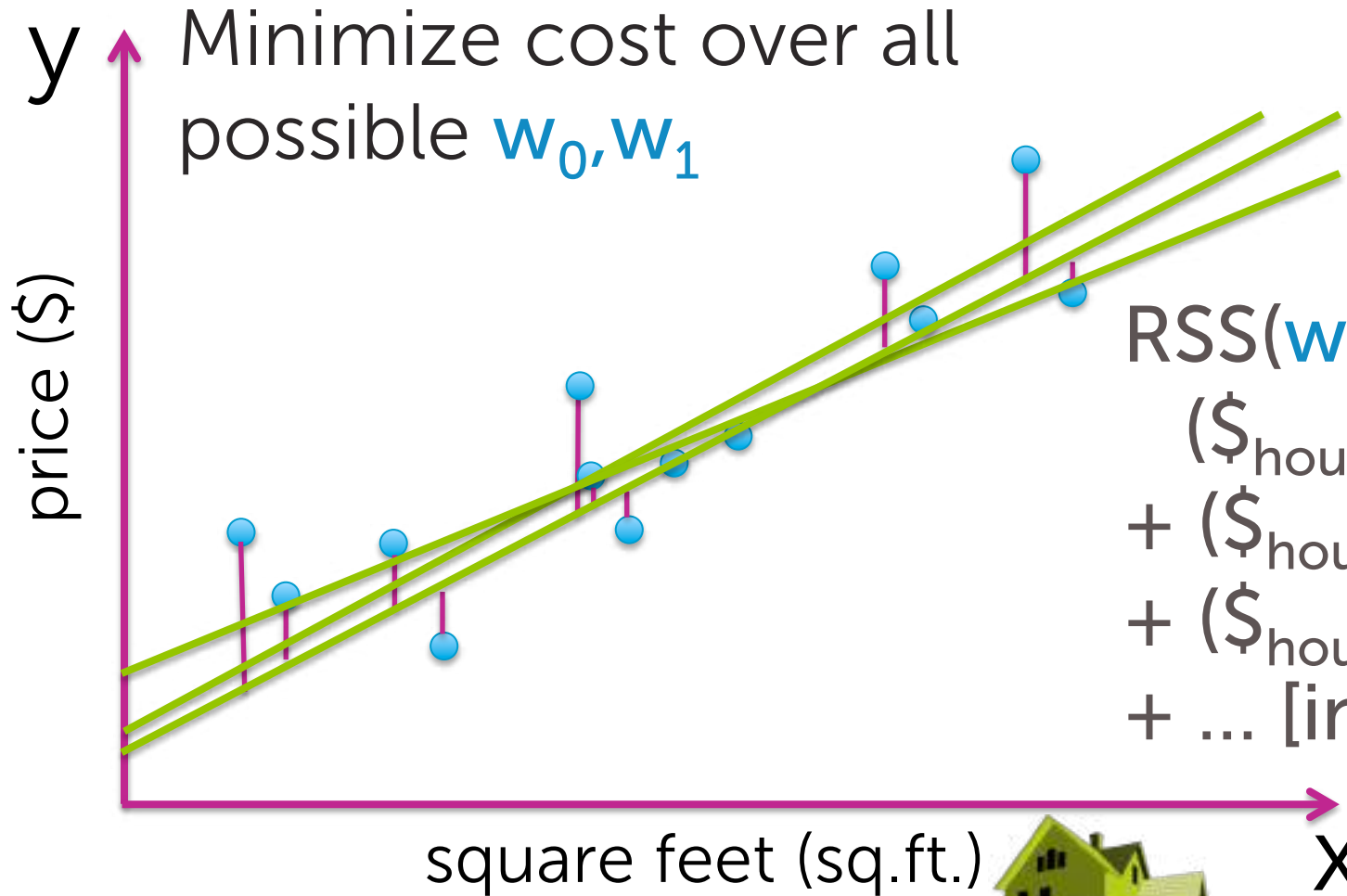


$$RSS(w_0, w_1) =$$

$$\begin{aligned} & (\$_{\text{house 1}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 1}}])^2 \\ & + (\$_{\text{house 2}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 2}}])^2 \\ & + (\$_{\text{house 3}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 3}}])^2 \\ & + \dots [\text{include all houses}] \end{aligned}$$



Find “best” line

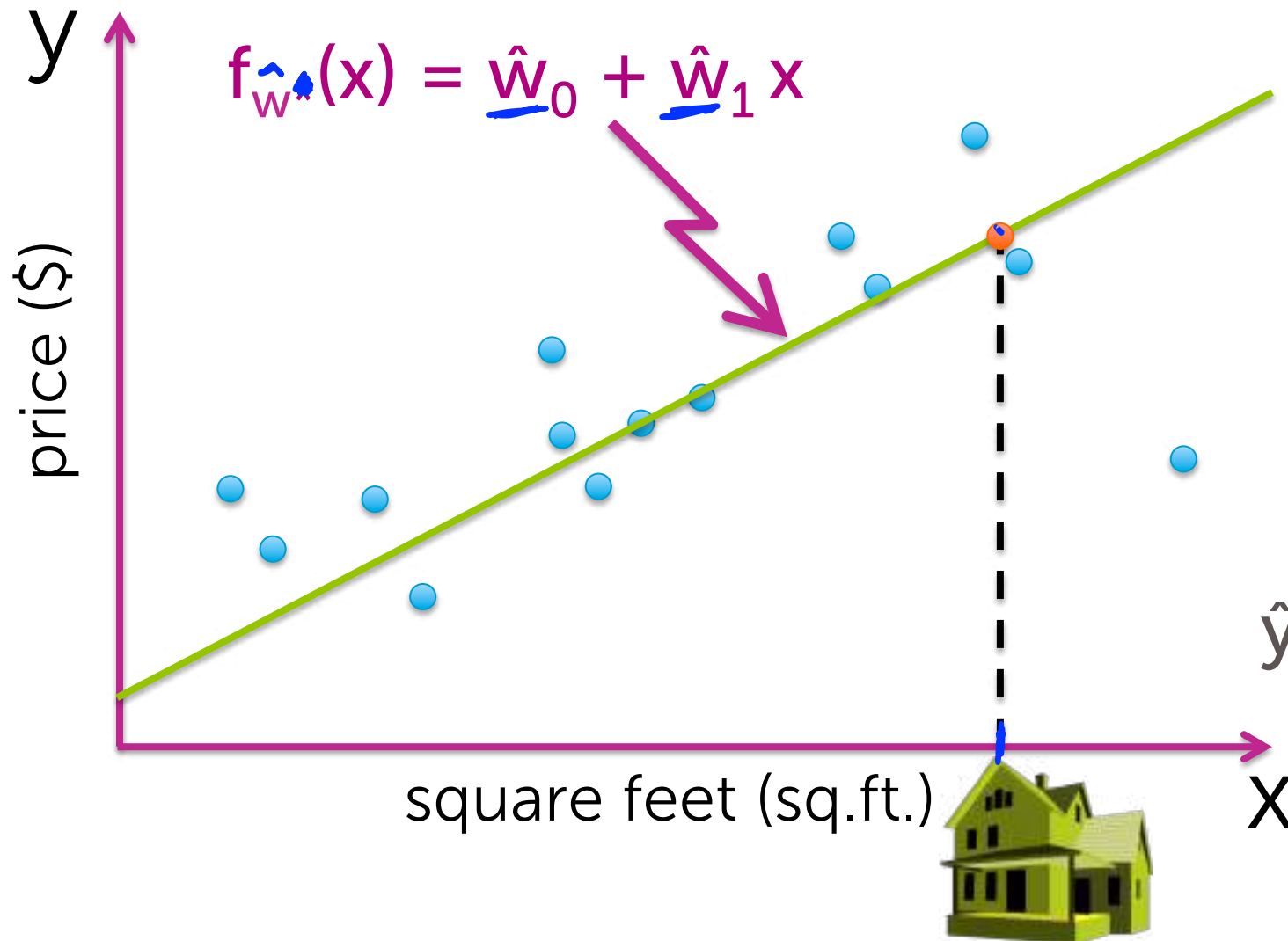


$$\begin{aligned} \text{RSS}(w_0, w_1) = & (\$_{\text{house 1}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 1}}])^2 \\ & + (\$_{\text{house 2}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 2}}])^2 \\ & + (\$_{\text{house 3}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 3}}])^2 \\ & + \dots [\text{include all houses}] \end{aligned}$$

↓

$$\hat{W} = (\hat{w}_0, \hat{w}_1)$$

Predicting your house price

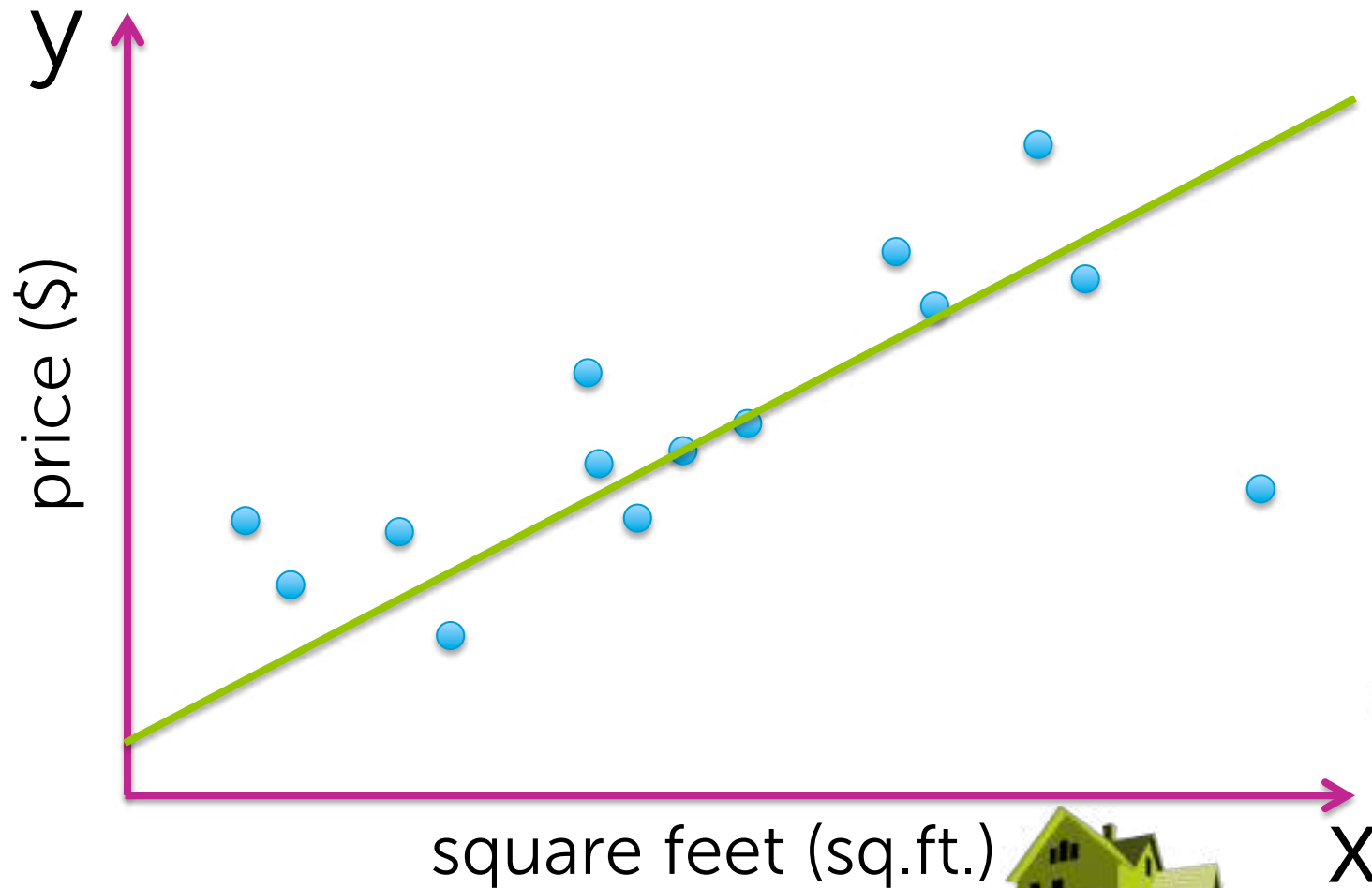


Best guess of your house price:

$$\hat{y} = \hat{w}_0 + \hat{w}_1 \text{sq.ft.}_{\text{your house}}$$

Adding higher order effects

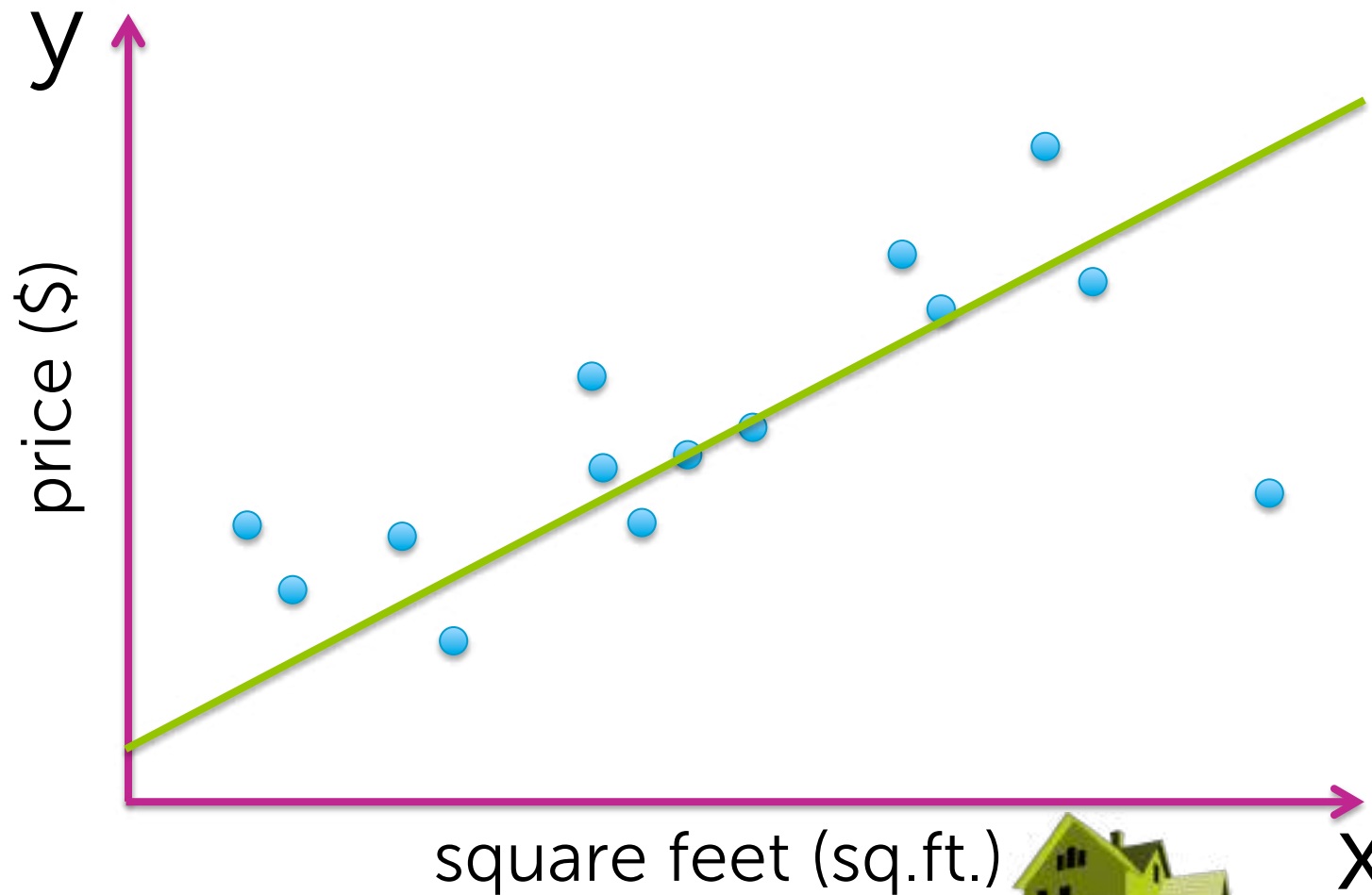
Fit data with a line or ... ?



You show
your friend
your analysis

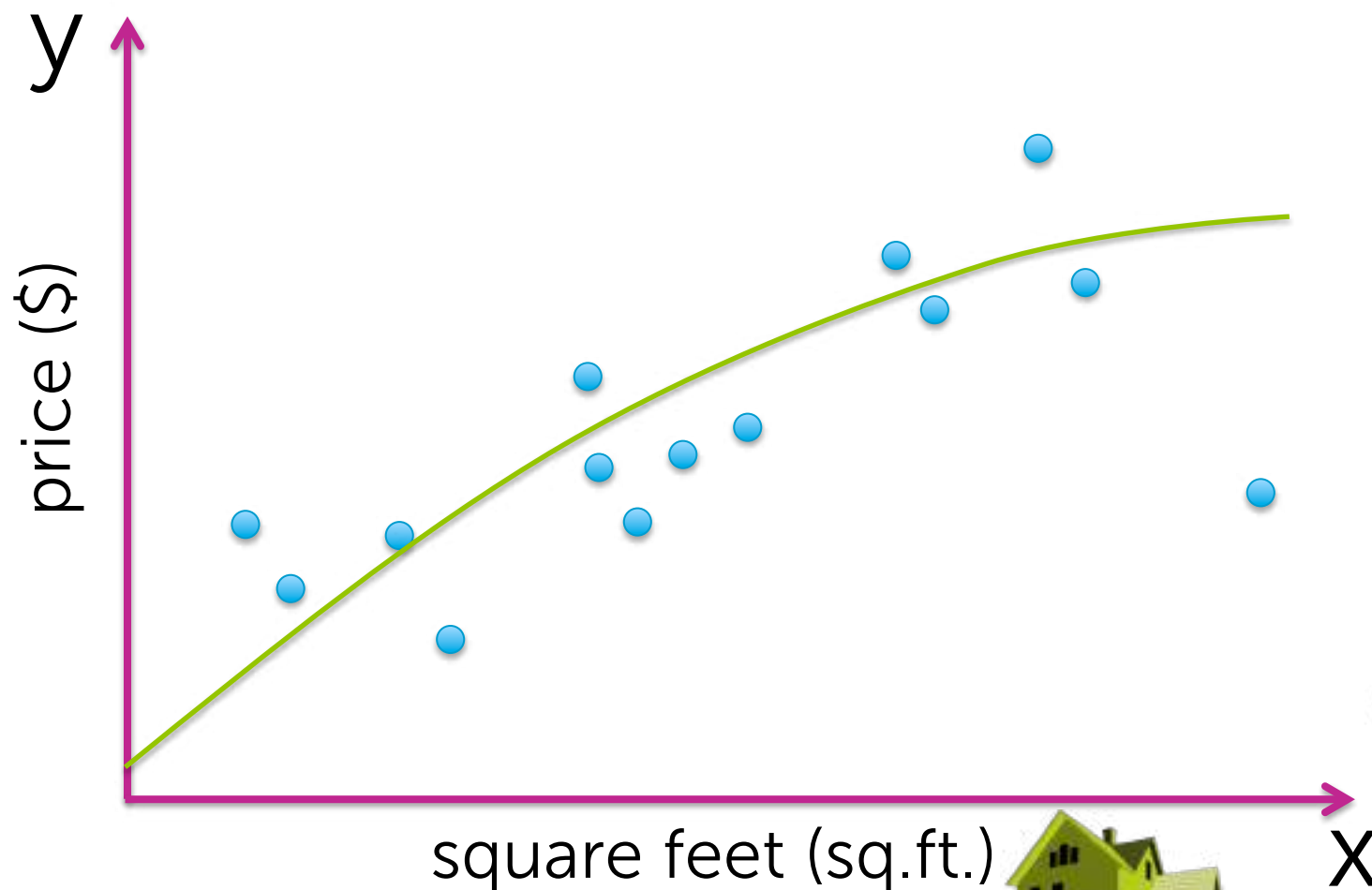


Fit data with a line or ... ?



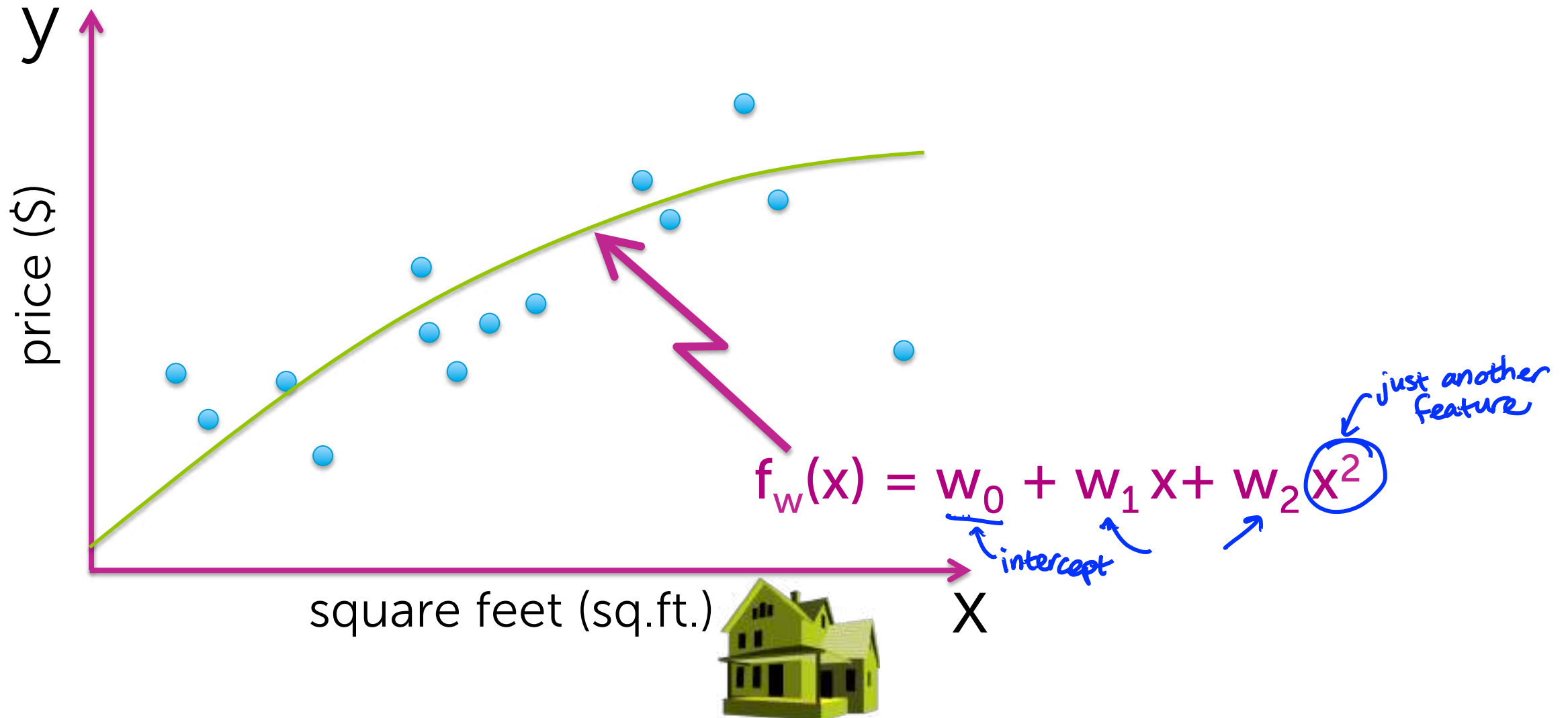
Bro
Dude, it's
not a linear
relationship!

What about a quadratic function?

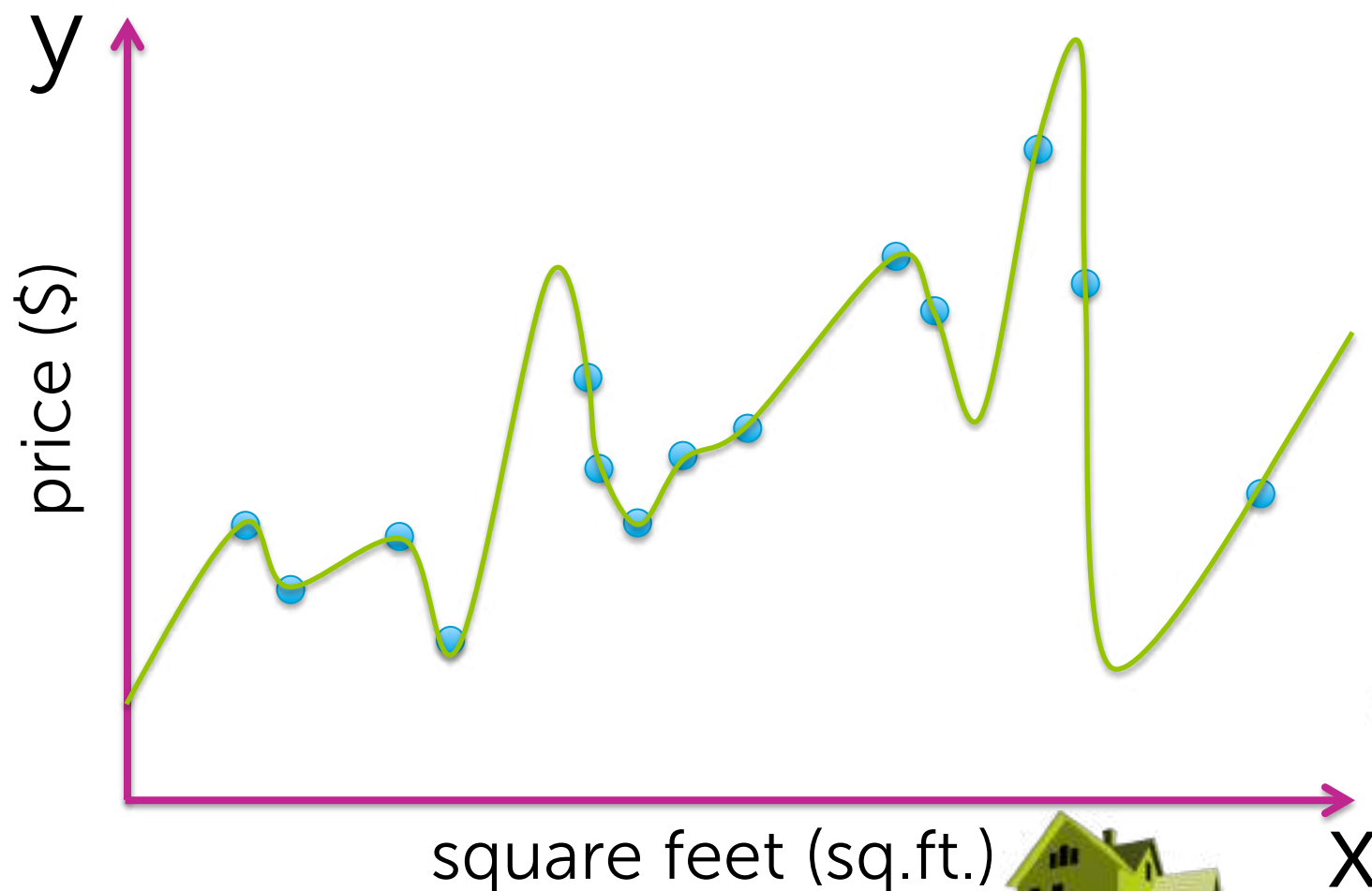


Dude, it's
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What about a quadratic function?

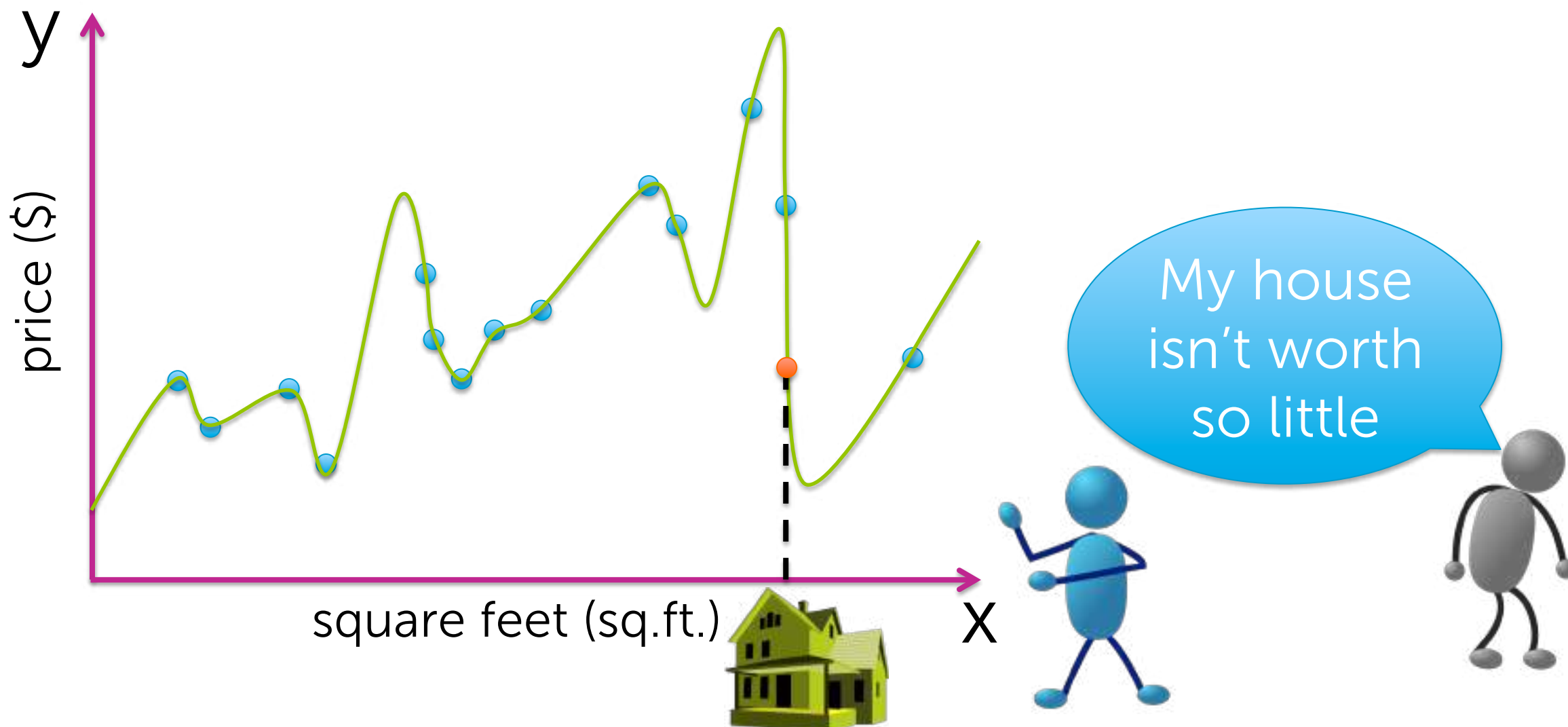


Even higher order polynomial



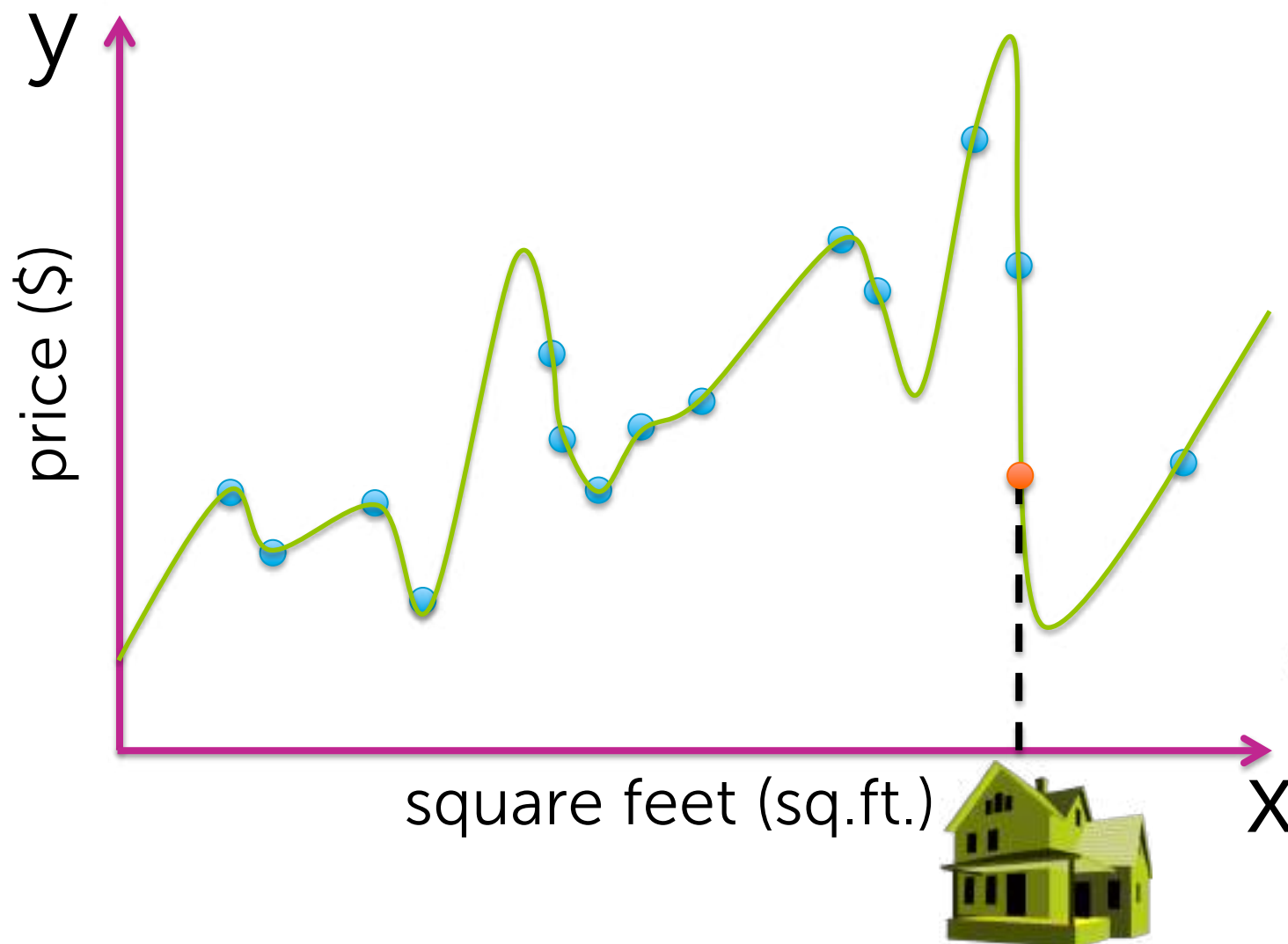
I can
minimize
your RSS

Do you believe this fit?



Evaluating overfitting via training/test split

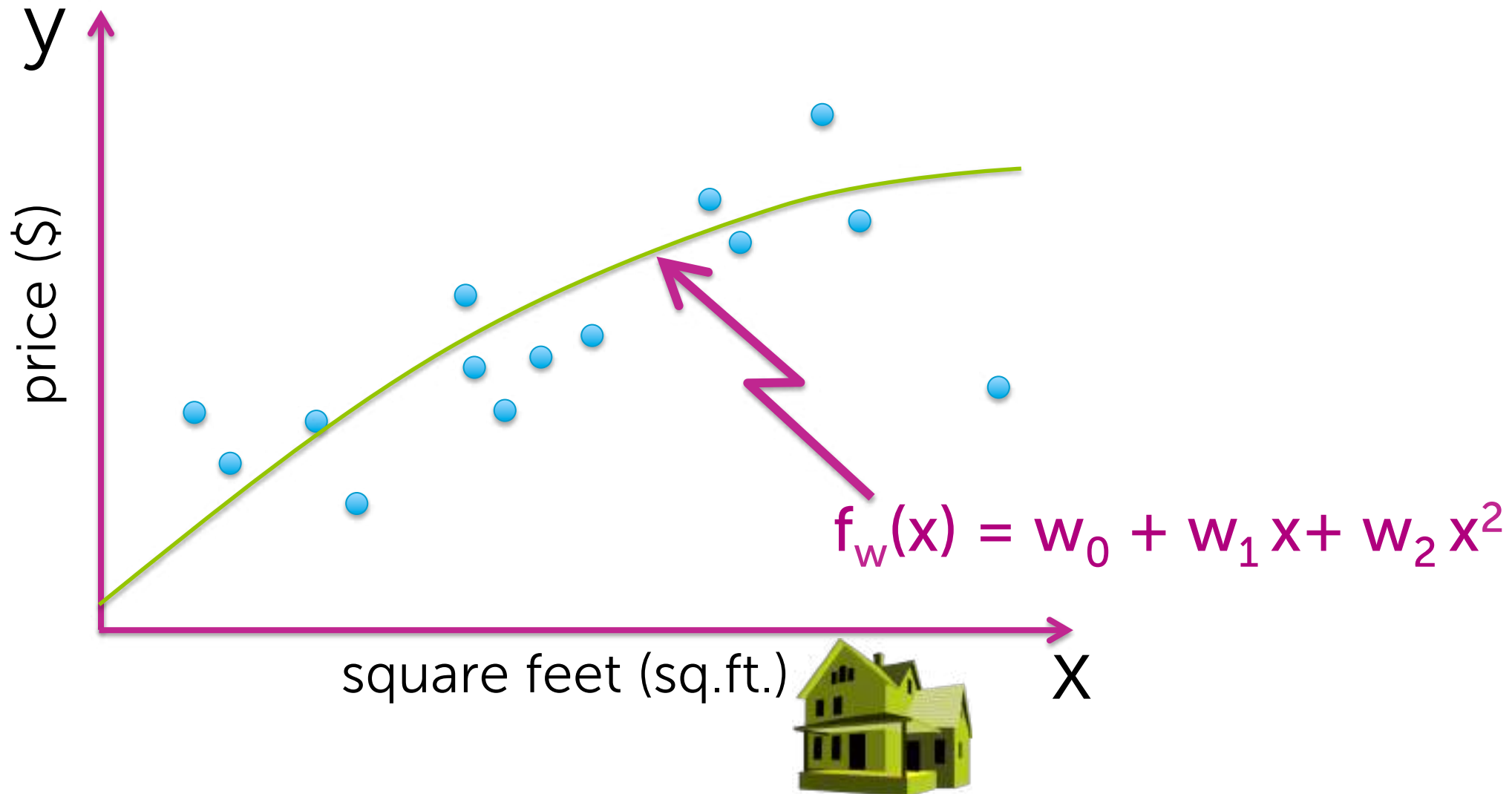
Do you believe this fit?



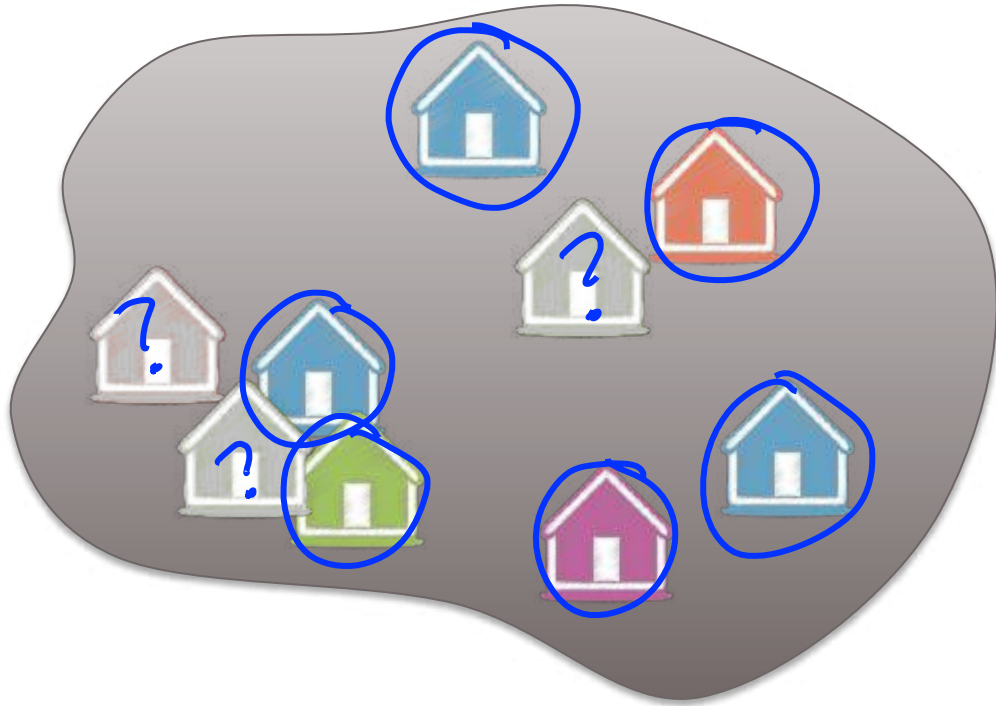
Minimizes RSS,
but bad predictions



What about a quadratic function?



How to choose model order/complexity



- Want good predictions, but can't observe future
- **Simulate predictions**
 1. Remove some houses
 2. Fit model on remaining
 3. Predict heldout houses

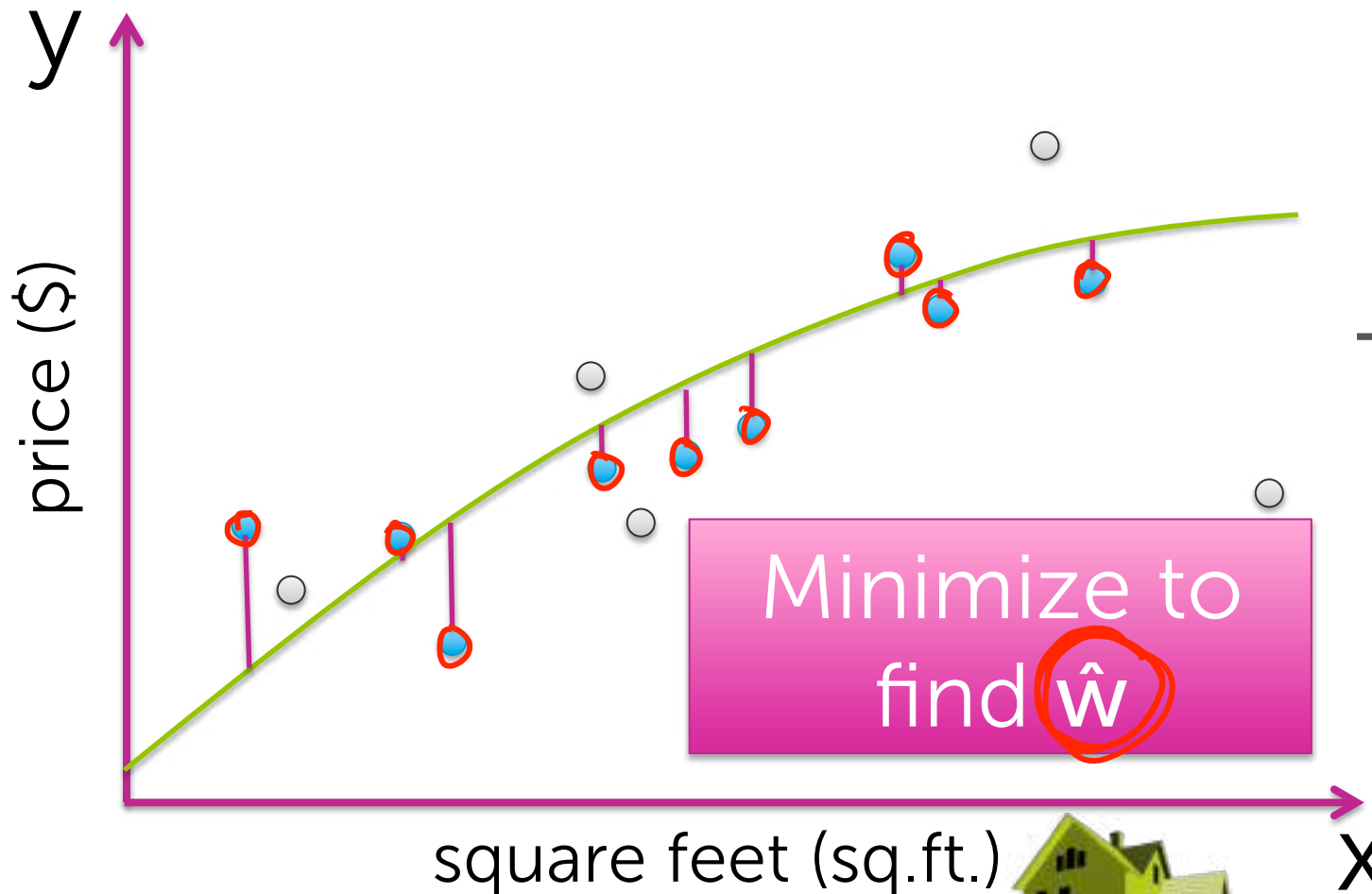
Training/test split



Terminology: – training set
– test set



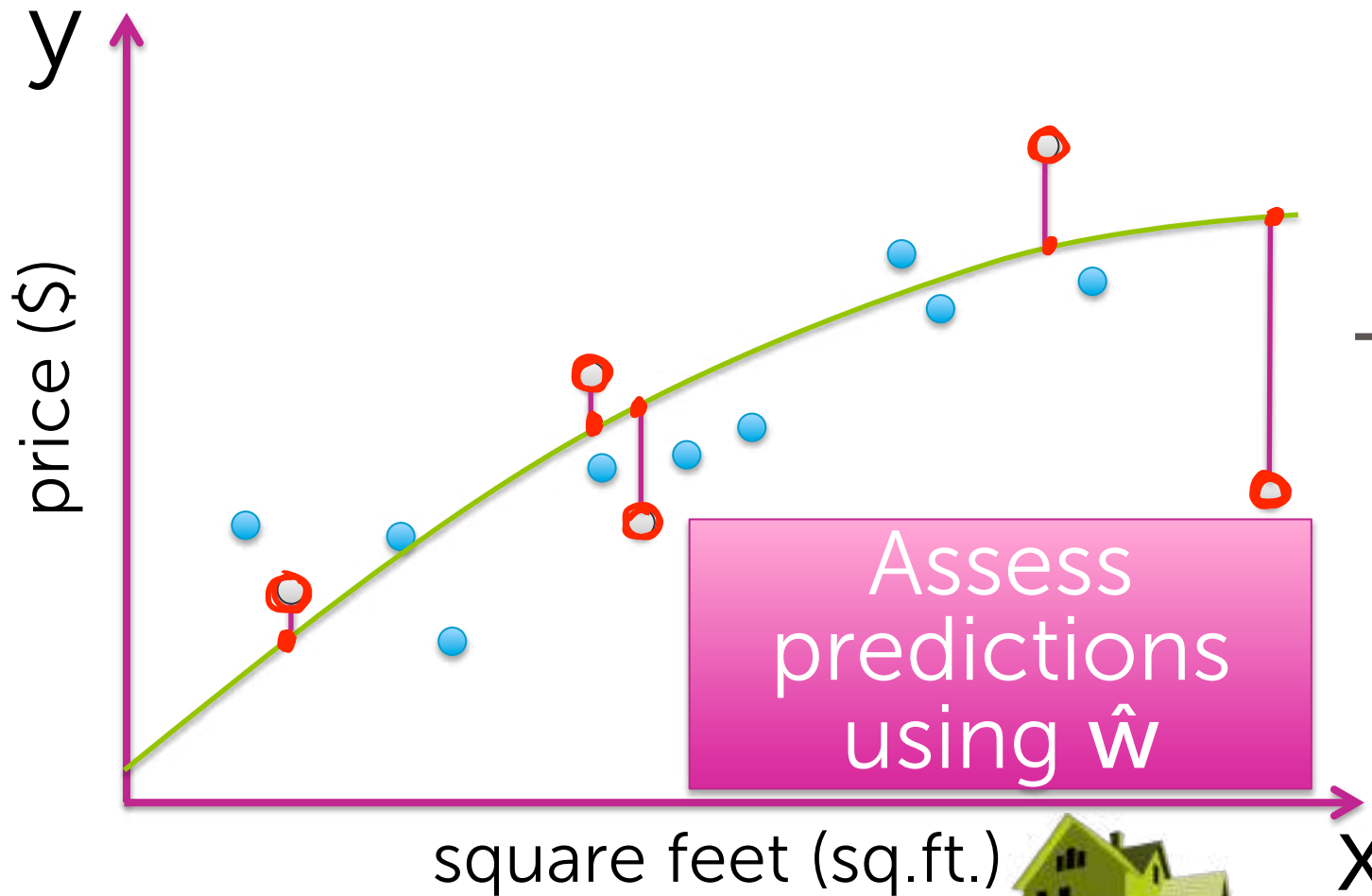
Training error



Training error (w) =

$$\begin{aligned} &(\$_{\text{train 1}} - f_w(\text{sq.ft.}_{\text{train 1}}))^2 \\ &+ (\$_{\text{train 2}} - f_w(\text{sq.ft.}_{\text{train 2}}))^2 \\ &+ (\$_{\text{train 3}} - f_w(\text{sq.ft.}_{\text{train 3}}))^2 \\ &+ \dots \text{ [include all} \\ &\quad \text{training houses]} \end{aligned}$$

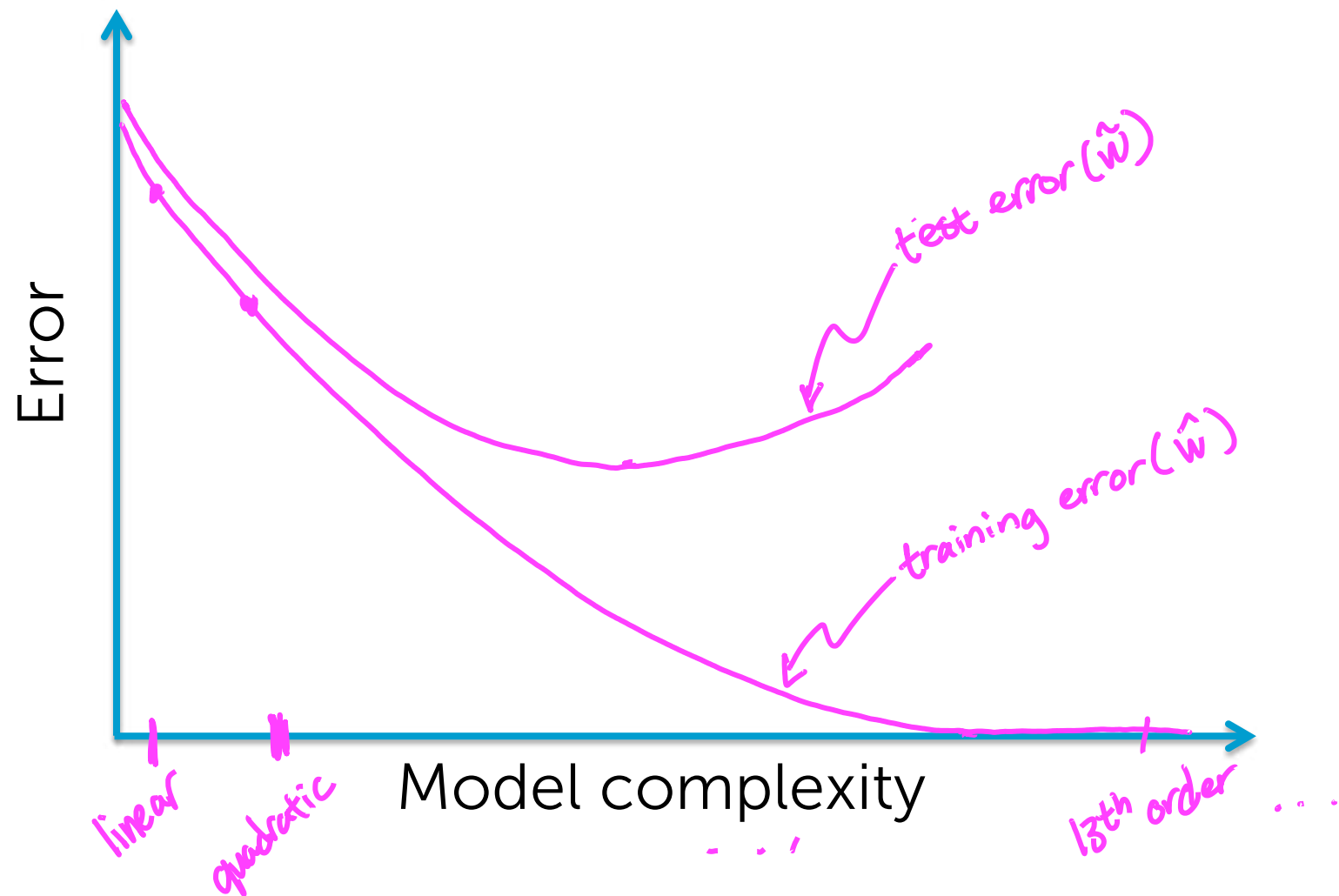
Test error



Test error $\hat{w}$ =

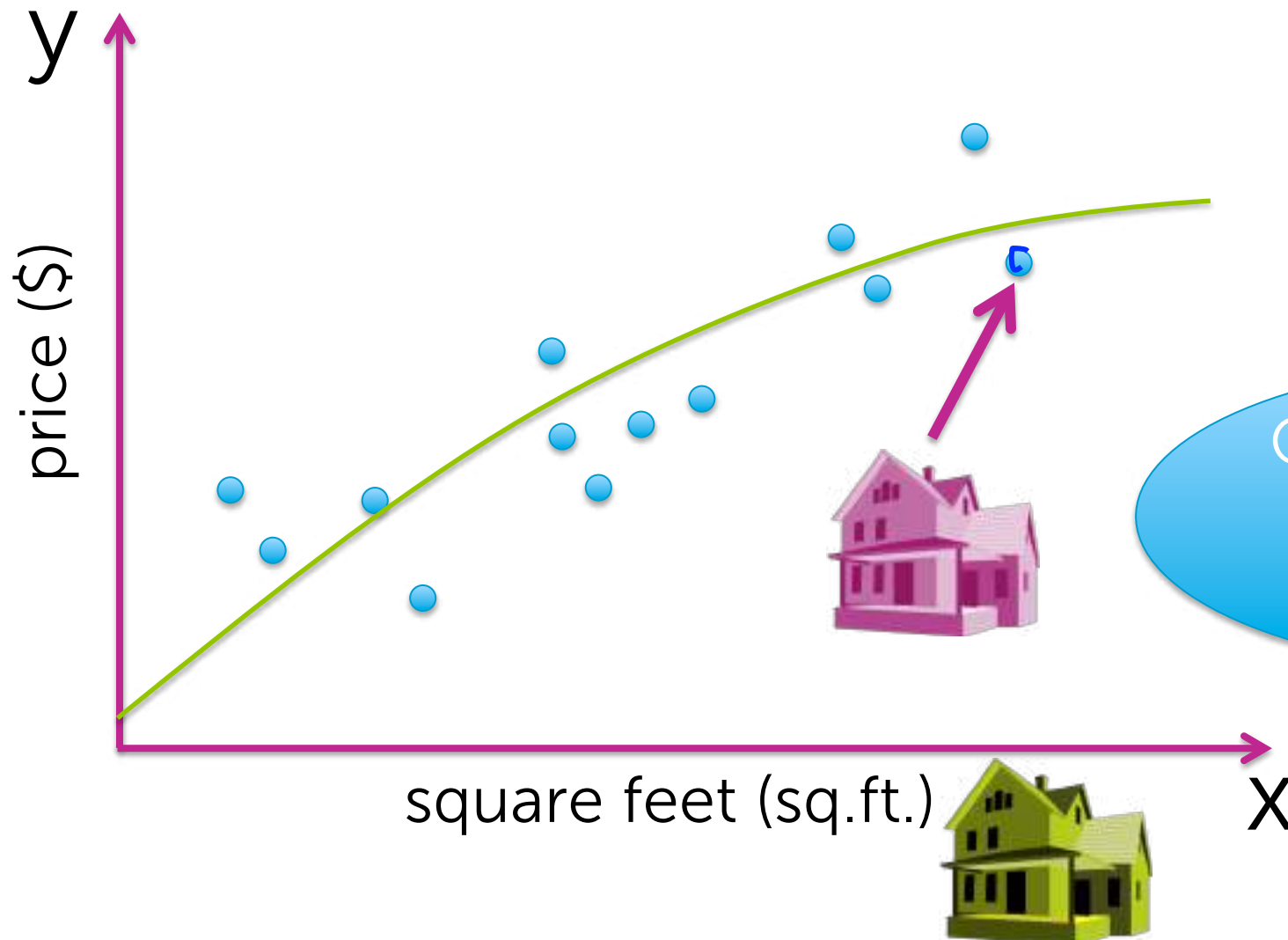
$$\begin{aligned} &(\$_{\text{test 1}} - f_{\hat{w}}(\text{sq.ft.}_{\text{test 1}}))^2 \\ &+ (\$_{\text{test 2}} - f_{\hat{w}}(\text{sq.ft.}_{\text{test 2}}))^2 \\ &+ (\$_{\text{test 3}} - f_{\hat{w}}(\text{sq.ft.}_{\text{test 3}}))^2 \\ &+ \dots \text{[include all test houses]} \end{aligned}$$

Training/Test Curves



Adding other features

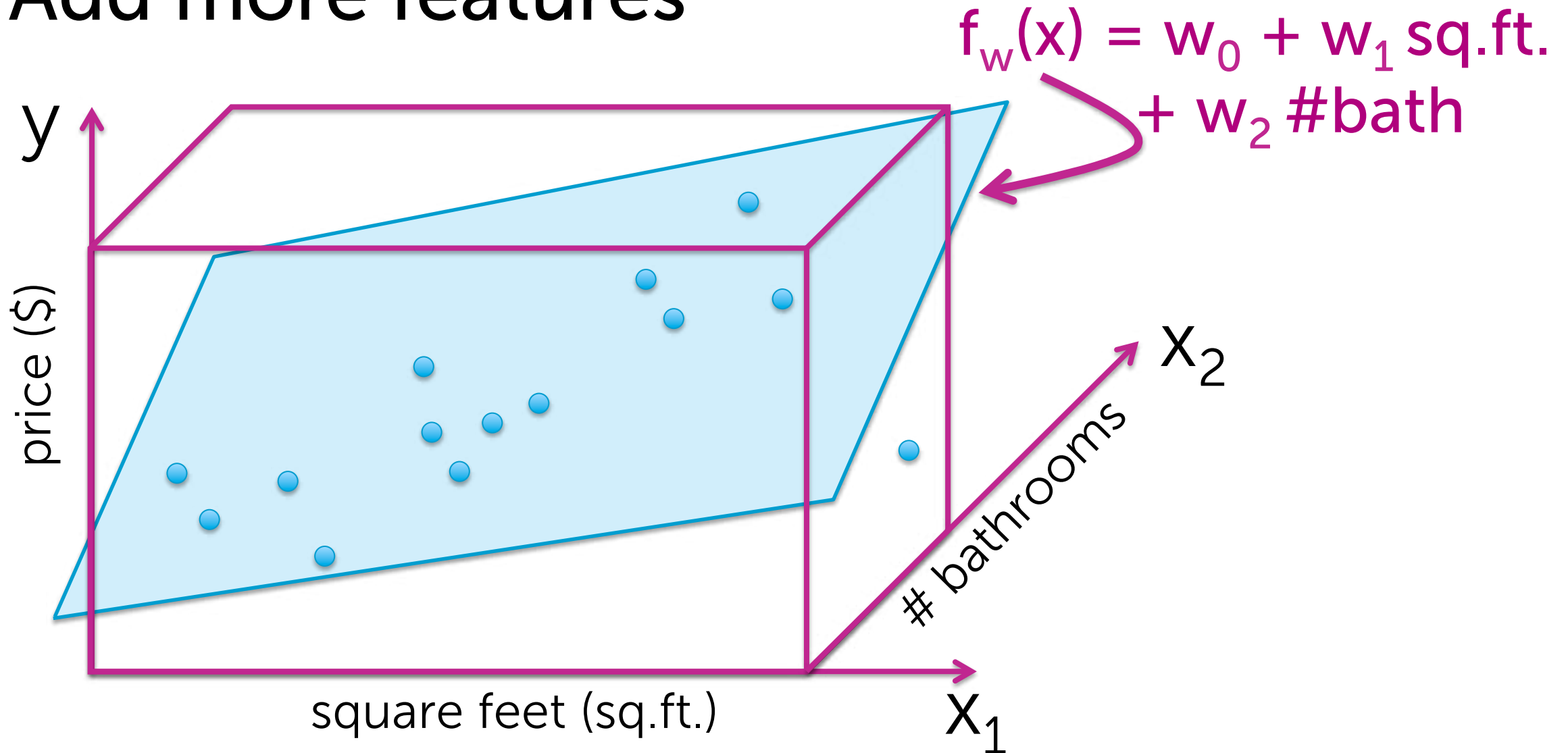
Predictions just based on house size



Only 1 bathroom!
Not same as my
3 bathrooms



Add more features



How many features to use?

- Possible choices:
 - Square feet
 - # bathrooms
 - # bedrooms
 - Lot size
 - Year built
 - ...
- **See Regression Course!**

Other regression examples

Salary after ML specialization



hard work



- How much will your salary be? ($y = \$\$$)
- Depends on x = performance in courses, quality of capstone project, # of forum responses, ...

Salary after ML specialization



hard work

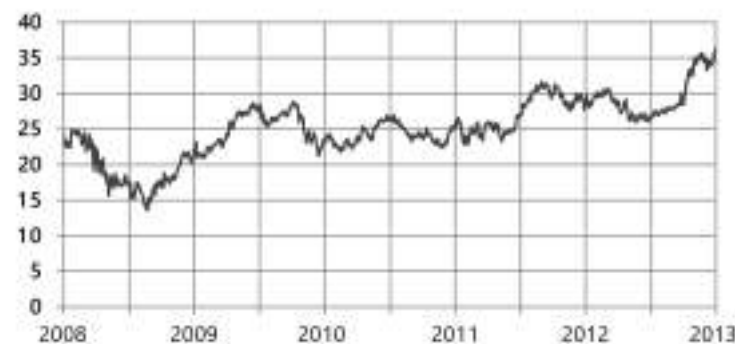


$$\hat{y} = \hat{w}_0 + \hat{w}_1 \text{ performance} + \hat{w}_2 \text{ capstone} + \hat{w}_3 \text{ forum}$$

informed by other students who completed specialization

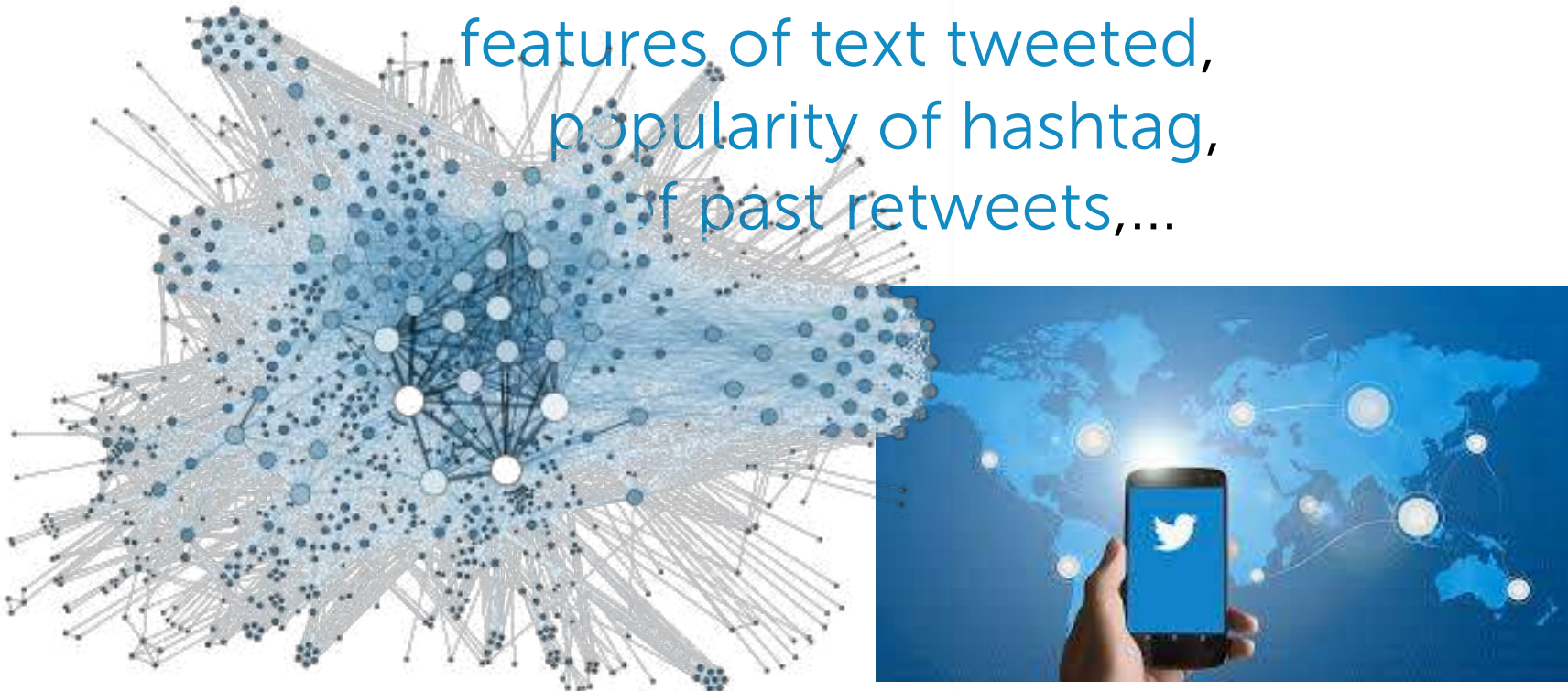
Stock prediction

- Predict the price of a stock
- Depends on
 - Recent history of stock price
 - News events
 - Related commodities



Tweet popularity

- How many people will retweet your tweet?
- Depends on # followers,
of followers of followers,
features of text tweeted,
popularity of hashtag,
of past retweets,...



Smart houses

- Smart houses have many distributed sensors
- What's the temperature at your desk? (no sensor)
 - Learn spatial function to predict temp
- Also depends on
 - Thermostat setting
 - Blinds open/closed or window tint
 - Vents
 - Temperature outside
 - Time of day



Summary for regression

What you can do now...

- Describe the input (features) and output (real-valued predictions) of a regression model
- Calculate a goodness-of-fit metric (e.g., RSS)
- Estimate model parameters by minimizing RSS (algorithms to come...)
- Exploit the estimated model to form predictions
- Perform a training/test split of the data
- Analyze performance of various regression models in terms of test error
- Use test error to avoid overfitting when selecting amongst candidate models
- Describe a regression model using multiple features
- Describe other applications where regression is useful